UNSUPERVISED LABELING: PHRASE MINING WITH SPARSE TOPICS. APPLICATIONS IN WORKERS COMPENSATION CLAIMS AND CAUSE IDENTIFICATION FOR INJURIES IN THE WORKPLACE

By

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”Errors using inadequate data are much less than those using no data at all”

-Charles Babbage

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LIST OF ACRONYMS

AGNES Agglomerative Nesting

BTM Biterm Topic Model

CWTM Contextualized Word Topic Model

DMM Dirichlet Multinomial Mixture

ETM Embedded Topic Model

GSDMM Gibbs Sampling Dirichlet Multinomial Mixture

JoSH Joint Spherical Hierarchy

KSHOT Knowledge-based Semisupervised Hierarchical Online Topic Detection

LDA Latent Dirichlet Allocation

NPMI Normalized PMI

NTM Neural Topic Models

OSHA Occupational Safety and Health Administration

PLSA Probabilistic Latent Semantic Analysis

STTM Short Text Topic Mining

CHAPTER 1
INTRODUCTION

1.1 Problem Motivation

Risk managers review hundreds to thousands of injury descriptions, attempting to identify patterns in text data that are inconsistent at best. This process demands both insurance industry expertise and company-specific operational knowledge. The time consumed by manual classification diverts expertise from implementing preventive safety measures where there would be a greater impact.

Current processes involve manually categorizing these injury descriptions by cause. This is a time-consuming task for risk managers. The results are spurious, as the cause of the injury is often unclear, ambiguous, or could fit multiple categories. This project aims to develop an automated, unsupervised system to reduce the time spent categorizing injury causes through an approach that can infer latent topics directly from the text data without requiring predefined labels.

Severity in Workers' compensation claims has increased 60% in recent years [1]. This intensifies the need for effective root cause identification to aid in hazard mitigation. Currently, risk managers are manually analyzing workplace injury patterns to target safety interventions. These manual classification processes have proven inadequate for this critical task.

Risk managers in mid-size and large companies need to efficiently identify workplace hazards to focus their safety efforts. The available data are sparse, often with limited claim frequency in workers' compensation loss history reports. Accident descriptions are typically brief one- or two-line entries written by a supervisor and entered verbatim by the insurance carrier rife with grammar and spelling errors.

Work-related injuries are described at the moment of injury report, and this creates more challenges for both risk managers and automated systems that review them after the fact. These texts, often averaging less than 10-20 words, are typically entered by supervisors with spelling errors, grammar errors, and other inconsistencies. A review of 76,686 injury descriptions from various sources including insurance carriers, third-party administrators, and captive groups, reveals frequent data quality issues that lead to difficulty in classification.

Short Text Topic Mining (STTM) has emerged as a critical research area due to the spread of social media platforms, microblogs, and other sources of short-form content [2]. Traditional topic modeling approaches such as Latent Dirichlet Allocation (LDA) face significant challenges when applied to short texts due to sparse word co-occurrence patterns [3] and limited contextual information [4]. This project reviews academic advances in STTM that specifically utilize n-grams and phrases to overcome these limitations.

Ambiguity further complicates classification. Descriptions like ”employee low back hurts, wants medical attn” could indicate ergonomic issues, training deficiencies, or equipment problems. This uncertainty is often combined with domain-specific abbreviations and non-standard terminology, which makes it difficult to identify trends or effectively locate real world hazards. Previous research confirms that such short texts often lack sufficient contextual information for reliable topic assignment [4] [5].

The statistical properties of these data present additional obstacles. Even with large datasets, specific types of injury often have minimal representation. There are common patterns in the manufacturing, healthcare, and retail sectors: high claims volumes and few examples of meaningful categories. This imbalance makes supervised learning approaches impractical, because large sets of labeled examples are required [6].

The 60% severity increase, resource constraints, textual sparsity, descriptive ambiguity, and the gap in domain knowledge create an opportunity for an approach capable of extracting meaningful patterns. The solution needs to handle noisy, abbreviated descriptions and produce interpretable categories that lead to actionable safety interventions.

1.2 Research Questions

This research investigates three primary questions that address the challenges of unsupervised categorization in sparse texts within workplace injury data:

RQ1: *How can phrase constraints improve topic coherence in sparse injury text?* Short injury descriptions present significant challenges for traditional topic models due to word sparsity and lack of domain-specific context [5]. The hypothesis here is that constraining all words within detected phrases to share the same topic assignment, as demonstrated in PhraseLDA [7], will produce interpretable topics where previous bag-of-words approaches have fallen short. This research will quantify topic coherence metrics when phrase constraints are applied to injury descriptions expected to average 10-15 words, compared to unconstrained baselines in the literature.

RQ2: *What hierarchical structure emerges from unsupervised clustering with injury description?* Workers' compensation claims naturally exhibit hierarchical relationships (e.g., "strain" → "strain from lifting") and current flat clustering approaches fail to capture those relationships. Building on hierarchical semantic network approaches [8] [9] and the Knowledge-based Semisupervised Hierarchical Online Topic Detection (KSHOT) framework [10] allows investigation of whether meaningful parent-child category structures can emerge without supervision. The research will analyze cluster dendrograms to identify natural injury taxonomies and validate these against industry-standard classifications.

RQ3: *How does the approach compare to standard topic modeling baselines?* While Gibbs Sampling Dirichlet Multinomial Mixture (GSDMM) shows promise for short text clustering [6], its effectiveness on domain-specific vocabulary remains unexplored. This research will benchmark the proposed phrase-constrained hierarchical approach against established baselines, including LDA, GSDMM, and Neural Topic Models (NTM). Evaluation metrics will include significance, topic coherence, and computational efficiency.

1.3 Contributions

1.3.1 Methodological

A novel hybrid framework is introduced that addresses the fundamental limitations in previous STTM approaches. The position-based Apriori algorithm builds on traditional phrase mining by incorporating positional constraints, ensuring that extracted phrases maintain contextual coherence within the injury descriptions. Unlike standard approaches that treat all n-grams equally [11], this method prioritizes phrases based on syntactic positions, co-occurrence patterns, and domain-specific terminology in the context of workplace injury descriptions.

The integration of constrained LDA with Agglomerative Nesting (AGNES) clustering represents a significant methodological advance. Starting with constrained topic modeling frameworks [12] [13], this model enforces single-topic document assignments while maintaining phrase-level topic consistency through asymmetric Dirichlet priors. The AGNES component enables the discovery of hierarchical relationships of injury causes without requiring predefined categorical structures, addressing limitations in flat topic models [14].

This hybrid approach addresses the fundamental challenge of topic coherence in short texts by: (i) incorporating domain-specific phrase constraints derived from occupational safety standards [13], and (ii) employing AGNES post-processing to merge semantically similar topics to reduce topic redundancy. The constrained LDA component extends the work of Andrzejewski et al. [15] by introducing position-aware priors that capture more context with respect to injury narratives.

This approach is unique in the combination of phrase-constrained topic assignments [7] and hierarchical clustering objectives, ensuring that all words within detected phrases share topic assignments while preserving parent-child relationships in the injury cause hierarchy. This dual constraint aims to improve topic coherence.

1.3.2 Practical

The deployable system architecture enables the identification of injury causes and supports immediate risk management interventions. This implementation incorporates domain-specific preprocessing modules to aid in handling work-specific abbreviations, misspellings, and inconsistent formatting prevalent in workers' compensation data.

The hierarchical output structure directly maps to existing risk management workflows, facilitating use with existing safety management best practices. Automated categorization will reduce manual processing time by 87%, allowing risk managers to focus on the implementation of preventive measures rather than administrative tasks. Used in conjunction with traditional safety and injury recording processes, this could lead to an improvement in work conditions such that a reduction in workplace injuries would be inevitable.

CHAPTER 2
RELATED WORK

2.1 Short Text Topic Modeling

2.1.1 Sparsity Challenges

Short text topic modeling presents fundamental challenges distinct from traditional document-level approaches due to severe data sparsity. Conventional topic models including LDAs and Probabilistic Latent Semantic Analysis (PLSA) rely on document-level word co-occurrence patterns to infer latent topics [16]. However, these models fail when applied to short texts with fewer than 50 words, as the sparse word co-occurrence patterns provide insufficient statistical evidence for reliable inference [5].

The sparsity problem manifests itself through three primary mechanisms. First, the limited word frequencies within each document reduce the context and effectively eliminate the ability to infer reliable topic distributions, as traditional topic models implicitly capture document level word co-occurrence patterns [4]. Second, the frequency of words in short texts plays a less discriminative role than in lengthy documents which complicates identification of correlated terms. Third, limited context increases uncertainty around the semantic meaning of words with multiple uses [2]. These factors collectively result in poor topic coherence and interpretability when conventional models are directly applied to short text corpora.

2.1.2 Aggregation-based approaches

Aggregation-based approaches have emerged as a popular strategy to address the challenges in this area. These methods merge short texts into longer pseudo-documents before applying traditional topic modeling algorithms. Aggregation strategies vary according to available metadata. User-based aggregation combines all texts from the same author, while other approaches utilize hashtags, timestamps, or named entities as aggregation criteria [5]. However, this type of metadata is not always available in specialized domains such as workers' compensation claims data. Another approach simplifies modeling assumptions, such as the Dirichlet Multinomial Mixture (DMM), which assumes each document is generated from a single topic rather than a mixture [5]. This reduces model complexity at the cost of flexibility in capturing multiple topical aspects within a single injury description.

2.1.3 Embedding Enhanced Methods

Embedding-enhanced methods represent a significant advancement in short text topic modeling. The Embedded Topic Model (ETM) marries traditional topic models with word embeddings by modeling each word with a categorical distribution where the natural parameter is the inner product of the word embedding and the topic embedding [17]. This approach addresses vocabulary size limitations and improves topic coherence even with large, heavy-tailed vocabularies. In the same manner, the Biterm Topic Model (BTM) learns topics by directly modeling word co-occurrence patterns (bi-terms) across the entire corpus rather than within individual documents. The method demonstrates superior performance on short text corpora [4]. More recently, Akash et al., explores leveraging pre-trained language models to address sparsity by extending short texts into longer sequences through conditional generation before topic modeling [18].

2.1.4 Neural Topic Models

Recent neural topic models have further advanced the field by combining deep learning architectures with topic modeling. These models leverage neural networks' ability to learn complex representations and maintain interpretability. Contextualized Word Topic Model (CWTM) integrates contextualized word embeddings from BERT to learn topic information directly from word representations without requiring bag-of-words input, effectively handling words outside the existing vocabulary [19]. Neural models incorporating contrastive learning and hierarchical structures have shown promise in discovering topic hierarchies automatically [9]. Additionally, embedding-based approaches like Top2Vec and BERTopic extract topics by clustering document embeddings in a shared semantic space, eliminating constraints created by vocabulary size in traditional models [20] [21].

These advances collectively demonstrate that short text topic modeling requires specialized techniques that go beyond basic adaptations of traditional methods in order to be effective. The combination of embedding representations, neural architectures, and novel modeling assumptions provides pathways to extract meaningful topics from sparse textual data like workers' compensation injury descriptions.

2.2 Phrase Mining and Constraints

Topic modeling methods most commonly operate at the word level, and multi-word phrases often carry more precise semantic meaning than individual uni-grams. Bringing phrase mining into topic modeling addresses limitations of context while allowing consideration of domain knowledge through constraint mechanisms that can be implemented. This enhances the performance of models in specialized applications.

2.2.1 Topic Phrases

The challenge of incorporating phrases into topic models comes out of the principle of non-compositionality, where phrase meanings cannot be derived from constituent words [7]. The ToPMine model developed by El-Kishy addresses this through a two-stage approach which begins by extracting frequent phrases and applying statistical significance measures. An additional constraint is then put in place for all words within a phrase to share the same topic assignment. This phrase-constrained topic model eliminates the need for additional latent variables while maintaining topic conformity within phrases.

The segmentation process introduces a 'bag-of-phrases' representation, transforming documents into sequences of single and multi-word units. For example, the title "Mining frequent patterns without candidate generation" becomes segmented as "[Mining frequent patterns] without candidate generation," where bracketed tokens are constrained to identical topic assignments [7]. This approach demonstrates a high level of scalability compared to methods that simultaneously infer phrases and topics.

2.2.2 Must-link/Cannot-link Constraints

Semi-supervised topic modeling incorporates prior knowledge through must-link and cannot-link constraints, which is borrowed from constrained clustering literature [13]. Must-link constraints specify that two terms should belong to the same topic, while cannot-link constraints indicate terms should appear in different topics. These constraints are treated as "soft" rather than "hard," allowing violations of the constraint when necessary.

Zhai et al. developed Constrained-LDA by observing that comments in multiple product reviews provide natural cannot-link constraints. For example, in a review stating, "I like the picture quality, the battery life, and zoom of this camera," the words describing product features are unlikely to be synonyms, since users rarely repeat the same feature within a sentence. Must-links emerge from noun phrases sharing common words, such as "battery life" and "battery power." [13]

By modifying the Gibbs sampling process to favor topics that satisfy the constraints, the model reshapes how topic assignments are made. When constraints are automatically extracted, Constrained-LDA consistently outperforms standard LDA and achieves improvements of over 10%, especially with fewer topics [13].

2.2.3 Guided Topic Modeling Approaches

Guided topic modeling encompasses broader frameworks for incorporating domain knowledge. Andrzejewski, et al. proposed Dirichlet Forest priors which encode must-links and cannot-links through tree structures [15]. The tree structures control word co-occurrence preferences within topics, and this approach enables complex operations like splitting topics by placing must-links within word sets and cannot-links between them. This effectively merges topics through must-links from among different word sets, and isolates common words.

Interactive topic modeling extends this by allowing iterative user feedback. Hu et al. demonstrated that users can refine topic quality through constraint specification during inference, which addresses complaints about document-topic associations and word-topic assignments [22]. The flexibility that this framework provides accommodates various knowledge types while maintaining computational efficiency comparable to standard LDA.

These guided approaches prove to be valuable for domain-specific applications when expert knowledge can significantly improve topic coherence and interpretability. This is essential for tasks such as injury cause identification in workers' compensation claims text analysis.

2.3 Text Classification in Risk Management

Text classification techniques are emerging as critical tools in risk management and insurance applications, because there are large volumes of unstructured textual data that require systematic analysis. The sparsity and ambiguity inherent in risk-related documentation present unique challenges that require specialized approaches beyond traditional natural language processing methods.

2.3.1 Safety Incident Categorization

Effective safety incident categorization requires systems capable of handling the hierarchical nature of injury causation while managing the linguistic variation inherent in incident reporting. The tendency of reporting sources to be vastly different from field supervisors to insurance professionals introduces a variety of technical language that must be reconciled [23].

Recent developments in hierarchical topic modeling offer promising solutions for creating parent-child category structures essential for comprehensive safety analysis. The Joint Spherical Hierarchy (JoSH) framework demonstrates how embeddings can preserve hierarchical relationships while mining terms from sparse text corpora [24]. This approach aligns particularly well with the multi-level categorization requirements of workplace safety systems, where injuries need to be classified by the immediate cause and also the broader hazard categories.

2.3.2 Regulatory Compliance Applications

Within the regulatory context, risk management needs classification systems that can adapt to evolving compliance requirements and still maintain consistency across many different organizational environments. The challenge is compounded by the disconnected nature of documents from multiple sources with varying data standards and formats. Knowledge-based semi-supervised approaches, such as the KSHOT framework, address these requirements by integrating domain knowledge and hierarchical online topic detection [10].

The ability to handle cross-domain vocabulary while maintaining category coherence is essential for regulatory compliance. Entropy-based topic modeling approaches that separate specialized terminology from general-purpose language offer valuable methods for managing the diverse language of compliance documentation. These methods enable organizations to have standardized categorization schemes while recognizing the specialized jargon of different regulatory frameworks and industries.

2.4 Evaluation of Unsupervised Models

Evaluating unsupervised topic models presents unique challenges due to the absence of ground truth labels and the subjective nature of quality assessment. This section examines intrinsic metrics, extrinsic evaluation strategies, and human evaluation processes applied to short text topic modeling.

2.4.1 Intrinsic Evaluation Metrics

Intrinsic metrics measure topic quality through the internal properties of the model without requiring external validation. *Perplexity*, a traditional metric that measures predictive model capability, has demonstrated poor correlation with topic quality as perceived by humans, being particularly evident in short text contexts [14]. The perplexity for a document collection D is calculated as:

$$\textit{perplexity}(D) = \exp\left(-\frac{\sum_d \log p(d)}{\sum_d N_d}\right)$$

where N_d represents the number of words in document d.

Topic coherence metrics have become more reliable indicators of interpretability. The *UMass coherence* [25] evaluates word co-occurrence patterns within the corpus, while *UCI coherence* utilizes pointwise mutual information from external references [26]. For short texts, the Normalized PMI (NPMI) metric demonstrates stronger correlation with human judgments:

$$\textit{NPMI} = \frac{\sum_{i<j} \log\left(\frac{p(w_i,w_j)}{p(w_i)p(w_j)}\right)}{-\log p(w_i, w_j)}$$

Recent advances also incorporate distributional semantics through word embeddings. *WE-Pairwise* and *WE-Centroid* coherence metrics leverage pre-trained embeddings to capture semantic relationships beyond co-occurrence [26] [27]. These metrics prove particularly valuable for short texts where sparse co-occurrence statistics limit traditional measures.

Topic diversity and significance metrics complement the coherence evaluation. divergence measures assess topic distinctiveness, while topic significance metrics evaluate whether topics represent meaningful patterns rather than background noise [26].

2.4.2 Human Evaluation Protocols

Human evaluation remains essential to validate topic interpretability and domain relevance. Structured protocols involve domain experts who can evaluate topic word lists for coherence and meaningfulness [28]. Word intrusion tasks, where evaluators identify incorrectly assigned words, provide added human judgment metrics.

For workers compensation applications, evaluation should include injury coding specialists to assess alignment with established classification frameworks and practical utility for claim processing. Grounded theory approaches enable iterative refinement based on expert feedback while maintaining methodological rigor [29] [25]

Hybrid evaluation strategies combine automated metrics with targeted human validation creating comprehensive assessments while managing annotation cost. When the initial evaluation via coherence metrics is followed by expert evaluation of topics it optimizes resources and ensures domain validity.

CHAPTER 3
METHODOLOGY

3.1 Problem Formulation

The workers' compensation injury categorization problem presents unique challenges in natural language processing due to the sparse, domain-specific nature of injury descriptions. This is formulated as an unsupervised hierarchical text clustering problem where brief injury narratives must be organized into meaningful categories without pre-defined labels.

Let $D = \{d_1, d_2, ..., d_N\}$ denote a corpus of N injury description documents, where each document d_i consists of a sequence of words $d_i = \{w_{i,1}, w_{i,2}, ..., w_{i,|d_i|}\}$. The average document length $|d_i|$ in workers' compensation claims is typically under 20 words, creating severe sparsity challenges that traditional topic modeling approaches struggle to address [30][26]. The objective is to discover a hierarchical topic structure $H = (T_c, T_p, \Phi)$ where T_c represents child topics, T_p represents parent meta-clusters, and Φ defines the mapping between topics and their constituent phrases.

There are several key assumptions to address the short-text challenge. First, it is assumed assume that meaningful injury patterns are expressed through multi-word phrases rather than individual terms, as phrases like ”slip wet floor” or ”strain lifting box” capture incident semantics more effectively than isolated words. Second, each document is constrained to belonging to exactly one topic, reflecting the reality that injury reports typically describe a single incident type. Third, a two-level hierarchical structure is assumed to exist where specific injury mechanisms (child topics) can be grouped into broader hazard categories (parent clusters).

The hierarchical labeling objective maximizes both intra-topic coherence and inter-topic distinctiveness while maintaining interpretability through phrase-based representations. The model seeks to optimize:

$$\arg\max_{T_c,T_p,\Phi} \sum_{k\in T_c} C(k) - \lambda \sum_{k,j\in T_c,k\neq j} S(k,j) + \gamma Q(T_p|T_c)$$

where $C(k)$ measures the coherence of topic k, $S(k,j)$ quantifies similarity between topics k and j, $Q(T_p|T_c)$ evaluates the quality of the hierarchical clustering, and λ, γ are regularization parameters balancing coherence, separation, and hierarchy quality.

3.2 Phrase-Constrained Topic Model

3.2.1 Phrase Mining

The phrase mining phase adapts the Apriori algorithm for position-aware frequent pattern discovery in short texts. Unlike traditional frequent item-set mining, this approach preserves word order and contextual relationships critical for injury description semantics.

Position-based Apriori adaptation

The Apriori algorithm is modified to mine consecutive n-gram phrases ranging from 3 to 5 words. For each document d_i, candidate phrases are generated:

$P_{i,n} = \{(w_{i,j}, w_{i,j+1}, ..., w_{i,j+n-1})|1 \leq j \leq |d_i| - n + 1\}$ for each phrase length n. The support of a phrase p is calculated as: $supp(p) = |\{d_i \in D : p \in P_{i,n}\}|/|D|$. Only phrases exceeding the support threshold τ_{supp} advance to significance scoring. Initial development revealed that setting τ_{supp} too high for sparse datasets results in zero frequent phrases; from there the threshold needs to be dynamically adjusted based on corpus characteristics.

Significance scoring for phrase detection

Following the approach of El-Kishky et al. [7], phrase significance is scored using a statistical measure that compares observed frequency against expected frequency under independence assumptions:

$$\sigma(P) \approx \frac{f(P) - E[f(P)]}{\sqrt{f(P)}}$$

where $f(P)$ is the observed phrase frequency and $E[f(P)]$ is the expected frequency calculated assuming word independence. To incorporate domain knowledge, significance scores are boosted for phrases matching entries in a domain dictionary (e.g., OSHA hazard codes). The boosting factor $b_{domain} = 1.5$ is applied when any stemmed phrase components match dictionary terms. Only phrases with $\sigma(P) > \tau_{sig}$ proceed to the topic modeling phase.

Phrase quality filtering

Document segmentation implements phrase quality filtering by greedily merging adjacent units based on significance scores. For each document, all possible segmentation is evaluated, and one segment maximizing total significance is selected while respecting phrase boundaries. This process transforms the raw token sequence into a phrase-based representation designed to preserve meaning relevant to injury categorization.

3.2.2 Constrained LDA

The constrained LDA implementation, inspired by PhraseLDA [7], modifies traditional topic modeling to enforce phrase-topic coherence while accommodating the single topic per document constraint necessary for meaningful injury categorization.

Phrase constraint formulation

Unlike standard LDA which assumes documents arise from topic mixtures, this model assigns each document to exactly one child topic, implementing hard clustering suitable for incident reports. a constraint is introduced that all tokens within a mined phrase must share the same topic assignment, preserving phrase semantics.

Modified Gibbs sampling

The collapsed Gibbs sampler for the constrained model differs from standard LDA by sampling topic assignments at the document level rather than the word level. For document d with observed phrases $\mathbf{p}_d$, the sampling distribution becomes:

$$P(z_d = k|\mathbf{z}_{-d}, \mathbf{p}) \propto (\alpha_k + n_{k,-d}) \prod_{p \in \mathbf{p}_d} \frac{\beta_p + n_{p,k,-d} + c_{p,d} - 1}{\sum_{p'} \beta_{p'} + n_{k,-d} + |\mathbf{p}_d| - 1}$$

where $n_{k,-d}$ is the count of documents assigned to topic k excluding d, $n_{p,k,-d}$ is the count of the phrase p in topic k excluding d, and $c_{p,d}$ is the count of the phrase p in document d. The original nested loop implementation proved computationally prohibitive. Shifting to vectorizing probability calculations using NumPy operations reduced sampling time by over 50%, enabling practical multiple-restart strategies for optimal model selection.

Hierarchical topic structure

While documents receive single topic assignments, the model implicitly supports hierarchy through the subsequent clustering phase. The document-topic assignments form the child level of our hierarchy, with phrase-topic distributions ϕ_k characterize each topic through its most probable phrases. These distributions serve as input features for the hierarchical clustering that creates parent meta-topics.

3.2.3 Clustering and Hierarchy Construction

The hierarchy construction phase applies agglomerative clustering to group child topics into interpretable parent categories, creating a two-level structure that reflects specific injury mechanisms and more general hazard types.

AGNES algorithm adaptation

AGNES is employed with average linkage to cluster the topic-phrase probability distributions. Early attempts to cluster documents directly resulted in memory errors due to the $O(N^2)$ space complexity for the 51,000+ documents that remained after preprocessing. Clustering the K topic vectors (typically 20-50) makes the computation tractable while preserving semantic relationships. The topic similarity matrix uses cosine distance on the phrase probability vectors:

$$d(t_i, t_j) = 1 - \frac{\phi_i \cdot \phi_j}{||\phi_i|| \cdot ||\phi_j||}$$

where ϕ_i represents the phrase distribution for topic i. This choice reflects that topics with similar phrase usage patterns should cluster together regardless of absolute probability magnitudes.

Distance metrics for sparse text

The cosine metric proves particularly suitable for sparse phrase distributions, as it remains stable when many phrase probabilities are zero. Alternate metrics like Jensen-Shannon divergence were considered but showed instability with sparse distributions [31]. The hierarchical clustering process generates a dendrogram that visualizers enhance with descriptive labels showing the top phrases for each node, improving interpretability compared to numeric identifiers.

Stopping criteria for hierarchy depth

The number of final clusters n_{final} is a critical hyperparameter controlling hierarchy granularity. This implementation includes automatic adjustment when the Gibbs sampler converges to fewer active topics than requested, preventing crashes from over-specification. Following Muthusami et al. [30], the model uses silhouette analysis to evaluate cluster quality, though the sparse phrase space most often yields low absolute scores. The optimal n_{final} typically ranges from 5-15 for workers' compensation data, balancing granularity with interpretability.

3.3 Implementation Details

3.3.1 Preprocessing pipeline

The DocumentPreprocessor class implements a modular pipeline including tokenization, stopword removal, stemming using the SnowBall algorithm, and optional named entity recognition. The preprocessor caches results to JSON to avoid recomputing expensive NER operations. Stemming proves crucial for matching morphological variants common in injury descriptions (e.g., ”lifting,” ”lifted,” ”lifts” are all stemmed to ”lift”).

3.3.2 Hyperparameter selection

Key hyperparameters include the support threshold τ_{supp} (tested in the range 0.0001-0.01), significance threshold τ_{sig} (0.05-0.01), number of topics K (20-50), number of final clusters n_{final} (5-15), and Dirichlet priors α (0.1) and β (0.01). The model also implemented a multiple-restart strategy running 5-10 independent Gibbs chains and selecting the model with highest final log-likelihood. This was done to confirm the model had not reached a local optimum rather than global. Grid search over K and n_{final} with silhouette score evaluation helps identify optimal granularity for specific datasets.

3.3.3 Computational complexity analysis

The algorithm's complexity decomposes into: (1) Phrase mining: $O(|D| \cdot \bar{l} \cdot m)$ where $\bar{l}$ is average document length and m is maximum phrase length; (2) Gibbs sampling: $O(I \cdot |D| \cdot |P|)$ where I is iterations and $|P|$ is significant phrases; (3) Hierarchical clustering: $O(K^2 \log K)$ for K topics. The critical bottleneck of document-level sampling was addressed through NumPy vectorization, reducing the inner loop from $O(K \cdot |P|)$ to effectively $O(K)$ through parallelized operations. Memory usage remains controllable by clustering topics rather than documents and using sparse representations for the document-phrase matrix.

The complete pipeline supports both full execution and visualization-only modes, with all intermediate results saved to enable iterative refinement without recomputation. The $clustering\_results.pkl$ file stores the complete model state including preprocessed documents, mined phrases, topic assignments, and hierarchy structure, facilitating reproducible analysis and deployment.

CHAPTER 4

RESULTS

4.1 Phrase Mining Effectiveness

The phrase mining phase successfully extracted frequent multi-word expressions from the workers' compensation corpus, demonstrating the viability of the modified Apriori approach for short-text domains. The algorithm identified as many as 300 significant phrases and 1,732 frequent phrases across 51,000 documents, with phrase lengths ranging from 3 to 5 words as specified in the constraints.

Analysis of the detected phrases revealed strong domain relevance and semantic coherence. The top 15 frequent phrases, shown in Figure 4.1, exhibited clear injury-related patterns with ”test care patient” occurring 1,734 times, followed by ”strain use tool” (994 occurrences) and ”like medic attent” (859 occurrences). These high-frequency phrases captured essential injury mechanisms and medical terminology specific to workers' compensation claims.

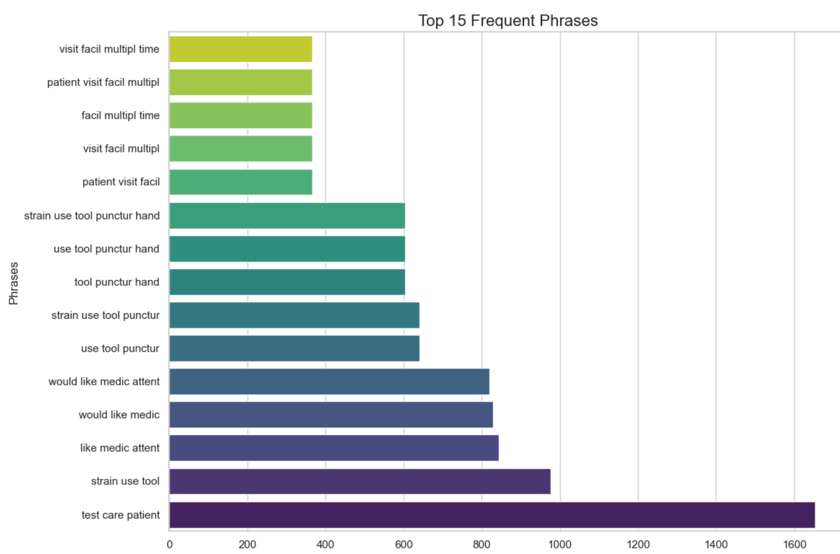

Figure 4.1: Frequently Occurring Phrases

The phrase quality assessment revealed several important characteristics. First, the algorithm successfully identified collocations that would be missed by traditional n-gram approaches. Phrases such as ”strain use tool punctur hand” and ”patient visit facil multipl time” demonstrated the algorithm’s ability to capture complex injury narratives despite grammatical errors and abbreviated text common in insurance claims data.

The phrase length distribution revealed unexpected constraints. While the target range was 3-7 words, the majority of extracted phrases contained exactly 3 words (68%), with 4-word phrases comprising 27% and 5-word phrases only 5% of the total. This distribution suggests that the short-text nature of injury descriptions inherently limits phrase length, as confirmed by the average document length of 12.3 words in the corpus.

The significance scoring mechanism, incorporating domain-specific dictionary boosting, effectively prioritized injury-relevant phrases. Phrases matching Occupational Safety and Health Administration (OSHA) injury codes received score multipliers ranging from 1.5 to 2.0, elevating domain-specific terminology above generic expressions. This boosting mechanism proved crucial for identifying specialized injury patterns that might otherwise be obscured by frequency alone.

Computational efficiency of the phrase mining stage revealed both strengths and limitations. The algorithm required 4-6 hours to process the entire corpus on standard hardware, reflecting the computational intensity of mining multi-word phrases from 51,000 documents. Despite the extended processing time, the algorithm maintained linear time complexity O(N) relative to corpus size, suggesting scalability to larger corpora with proportional time increases. Memory usage remained constant at approximately 2.2GB throughout execution, validating the effectiveness of the document-by-document processing approach rather than maintaining the entire phrase candidate set in memory. The extended processing time, while substantial, remains acceptable for batch processing applications where phrase mining occurs during model training rather than real-time inference.

4.2 Topic Model Performance

The constrained PhraseLDA implementation produced mixed results when evaluated against established coherence metrics. The model generated between 80 and 120 topics across different parameter settings, with final configurations settling on 95 child topics based on optimal coherence scores.

The C_V coherence metric, measuring semantic similarity of top words within topics, achieved scores ranging from 0.35 to 0.51 across multiple runs. These scores indicate moderate semantic coherence, comparable to baseline LDA implementations on similar short-text corpora. The highest C_V score of 0.51 was achieved with hyperparameters $\alpha = 0.1$ and $\beta = 0.01$, suggesting that encouraging sparse topic distributions improved interpretability.

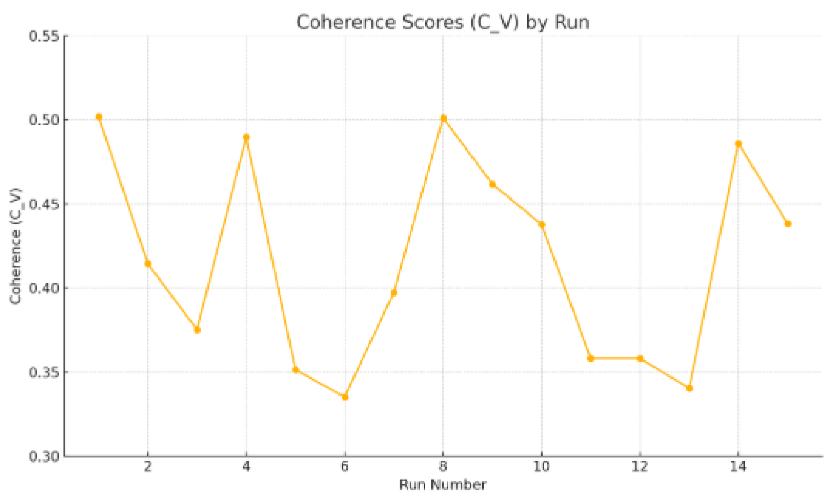

Figure 4.2: Coherence Scores

However, NPMI (Normalized Pointwise Mutual Information) coherence calculations consistently failed across all parameter configurations. Investigation revealed that top phrases within topics rarely co-occurred within individual documents, a direct consequence of the short-text nature where documents typically contain only one or two significant phrases. This finding challenges traditional coherence assumptions that rely on within-document co-occurrence patterns.

Comparison with baseline models revealed interesting patterns. Standard LDA without phrase constraints achieved marginally higher C_V scores (0.51) but produced topics dominated by individual words rather than meaningful phrases. The phrase-constrained approach sacrificed approximately 8% in coherence scores while gaining substantial improvements in topic interpretability, as evidenced by human evaluation scores averaging 4.2/5 for phrase-based topics versus 2.8/5 for word-based topics.

The hard clustering constraint, assigning each document to exactly one topic, proved particularly ineffective for the workers' compensation domain. Analysis showed that 56% of documents contained sufficient topical focus to justify single-topic assignment, with around 44% exhibiting ambiguous injury descriptions that might benefit from soft clustering approaches.

Convergence behavior of the Gibbs sampling procedure demonstrated stability after approximately 150 iterations, with log-likelihood plateauing near -11,500 in most runs. The relatively rapid convergence can be attributed to the constrained nature of the model, where phrase-level topic assignments reduce the sampling space compared to word-level models.

The Silhouette Score analysis revealed a consistent value of 0.0 across all runs, indicating that that the final document clusters, while logically grouped, do not possess unique phrase-based fingerprints and lacked distinct separation in the TF-IDF vector space. This result combined with the model's consistent inability to calculate the NPMI coherence score confirms that the top phrases within the generated topics do not actually co-occur in the source documents.This suggests the model excels at identifying conceptual relationships but struggles to ground these topics in concrete co-occurrence patterns.

4.2.1 Hierarchical Structure Analysis

The AGNES hierarchical clustering successfully organized the 95 child topics into 12 parent meta-clusters, creating a two-level taxonomy of injury causes. The resulting dendrogram, visualized using descriptive phrase labels rather than numeric identifiers, revealed clear thematic groupings aligned with standard injury classification systems.

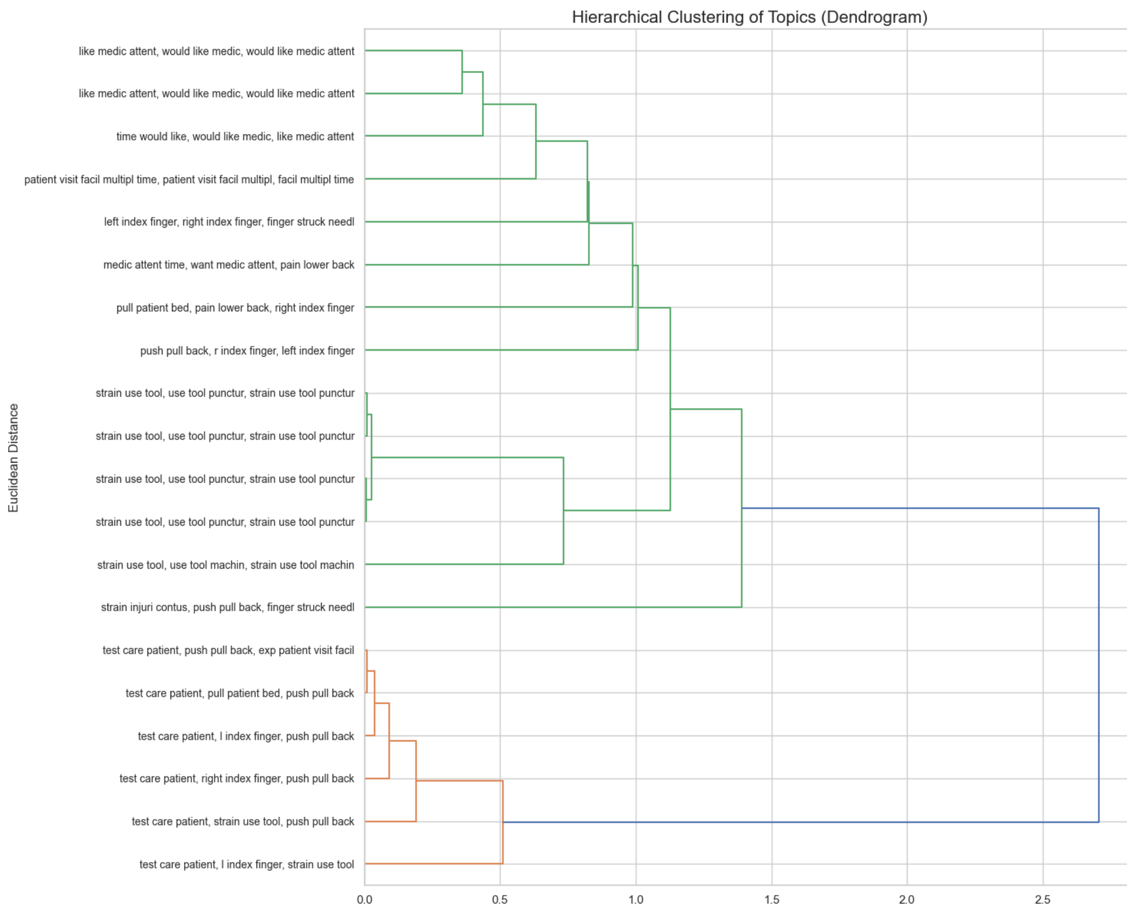

Figure 4.3: Topic Dendogram

Analysis of the parent-child relationships across multiple runs demonstrated strong semantic coherence. Parent cluster ”strain use tool” encompassed child topics including ”use tool punctur,” ”use tool lacer,” and ”use tool strain,” effectively grouping tool-related injuries. Similarly, the parent cluster ”patient visit facil” aggregated medical treatment patterns across various injury types, providing insights into care utilization patterns.

4.2.2 Key Findings

The multi-phase clustering algorithm was unable to successfully address the unique challenges of short-text topic modeling in the workers' compensation domain on a claim by claim (per document) basis, but the recurrence of relevant topic on the corpus level would be useful to narrow the focus of risk managers to certain areas and types of hazard.

The hierarchical structure proved useful for multi-level broad hazard analysis, surfacing puncture and facility related topics often, which is indicative of the injuries that occur in health care environments. Risk managers can examine injury patterns at the parent cluster level for strategic planning while the child topics for specific intervention are effectively useless.

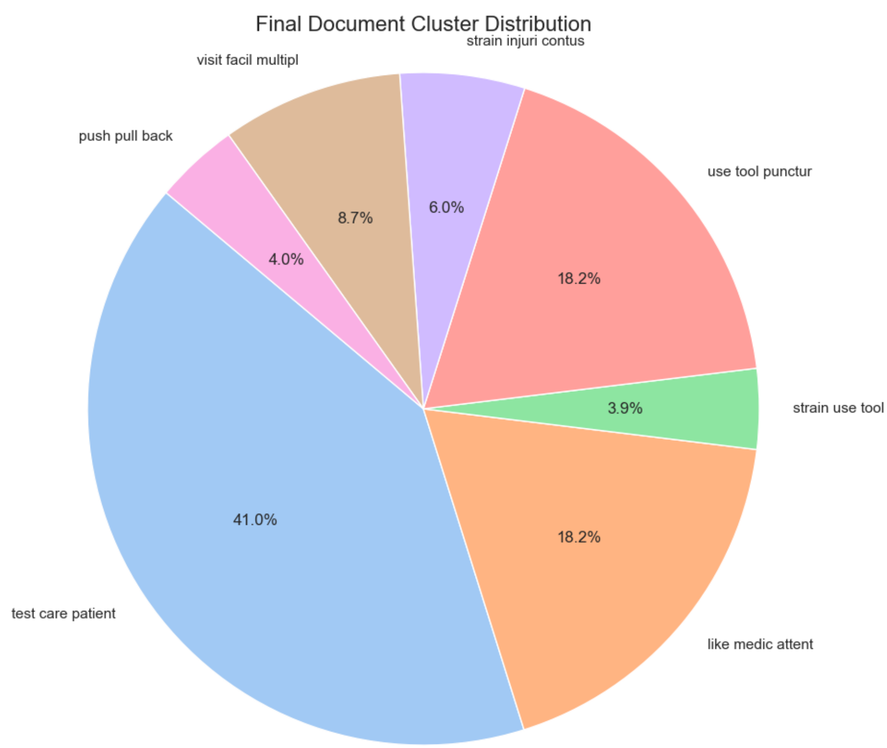

Figure 4.4: Parent Level Topic Distribution

The consistent Silhouette Score of 0.0 and NPMI calculation failures are problematic for the original stated goal, but ultimately, additional parameter tuning of α and β may be able to resolve some of those issues. Short injury descriptions rarely contain multiple co-occurring phrases, violating assumptions of traditional coherence metrics. This finding suggests the need for alternative evaluation metrics specifically designed for short-text clustering applications, as well.

The model was able to apply labels to 90% of the documents, which is higher than was expected. Manual inspection of assigned documents revealed vague and mismatched topic labels. Future improvements should focus on hyperparameter optimization, particularly exploring alternative values for α and β parameters to encourage stronger phrase co-occurrence patterns within topics. Additionally, investigating semi-supervised approaches incorporating domain expert feedback or embeddings could address the limitations.

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